

STAT120: Quiz 2 (9/10)

Name: _____

1. Which line of R code creates a vector containing the values 1, 4, 7, and 10?

☐ `c(1, 4, 7, 10)`

☐ `seq(from = 1, to = 10, by = 4)`

☐ `rep(c(1, 4), 7)`

☐ `1:10`

2. Suppose `v <- c(10, 20, 30, 40, 50)`. What is the output of `v[-2]`?

☐ 20

☐ 10, 30, 40, 50

☐ 10, 20, 30, 40

☐ An error message

3. Write R code that creates a vector called `x` containing the numbers 1 through 100.

4. Write R code that calculates the mean of the vector `x` from Question 3.

5. Suppose `gpa <- c(3.2, 3.8, 2.9, 3.9)`. Which line of R code returns the positions of the GPAs greater than 3.5 (i.e., should return 2 and 4)?

☐ `gpa(gpa > 3.5)`

☐ `which(gpa > 3.5)`

☐ `which[gpa > 3.5]`

☐ `gpa > 3.5`

6. Using the vector `gpa` from Question 5, write R code that prints the GPA values greater than 3.5 without manually typing their positions.

7. The `mtcars` data set has a variable called `am`, where 0 represents an automatic transmission and 1 represents a manual transmission. Write R code that prints the `mpg` values for cars with manual transmissions.

8. Using the `mtcars` data set, write R code that prints the car models whose `mpg` is less than 16. Assume that the car model names have been stored in a vector called `model`.